

Matys (Adéas) BIECHE

Seeking a 6-month final-year internship in autonomous robotics starting March 2027

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EDUCATION

- **ENSTA – IP Paris** – Engineering Degree in Autonomous Robotics; GPA 4.0/4.0 2025–2027 · Brest
- **SeaTech Toulon** – Engineering Degree in Mechatronic Systems, dual degree; GPA 4.0/4.0 2023–2027 · Toulon

EXPERIENCE

- **Enchanted Tools** : *R&D Robotics Engineer Intern – Humanoid Navigation* May–Sept. 2026 · Paris
 - **Semantic navigation**: End-to-end design and development of user-configurable zones (keep-out, speed-restricted, contextual), with a ROS 2 backend (C++/Python) integrated into Nav2 and a configuration interface (TypeScript/React, Qt). Shipped as a product feature.
 - **Relocalization**: Implemented absolute pose correction from visual landmarks detected by the perception system, complementing NVIDIA GPU-accelerated cuVSLAM and eliminating localization drift of up to 1 m.
 - **Benchmark**: Benchmarked the navigation and mapping stack and identified an MPPI controller failure mode, leading to a dedicated team effort.
 - **Demo**: Technical presentation of the robots to the public and industry professionals at VivaTech 2026 and ROSConFR 2026.
- **ENSTA** : *Robotics Teaching Assistant* Feb.–May 2026 · Brest
 - **Robotics platform**: Developed the ROS 2 stack for an educational robot: motor/encoder/camera drivers, microcontroller integration and serial/CAN/I2C/SPI communication, Docker/systemd deployment, and SSH access for line-following algorithm development.
 - **Teaching**: Taught and supervised lab sessions on UAVs and ROS 2.
- **Tokyo University of Science** : *Robotics R&D Intern* Apr.–Aug. 2025 · Tokyo
 - **Acoustics**: Designed the analog electronics and STM32 firmware for an underwater communication link for a robotic fish: amplification/filtering, impedance matching, ADC/DAC, SPI/DMA, and OOK modulation/demodulation; achieved 1 kb/s communication with €75–85 hardware cost.
 - **Embedded AI**: Developed an optical receiver on Raspberry Pi: LED detection/tracking using computer vision (OpenCV) and reconstruction of noisy sequences using a GRU network (PyTorch); 95.8 % accuracy, >80 % under severe simulated noise.

COMPETITIVE ROBOTICS

- **Robot Club de Toulon – RoboCup MSL** Toulon
Goalkeeper Robot Lead – 2026 World Runner-up 2023–2026
 - **Localization**: Developed in-game robot localization using camera–3D LiDAR SLAM fusion, ICP registration, and geometric field landmarks (goalposts), with explicit dynamic-object rejection.
 - **Stereo vision**: Designed a modified and synchronized pair of 1080p JeVois Pro cameras (40 cm baseline) for 3D ball localization beyond 20 m with Kalman-filter trajectory prediction.
 - **Goalkeeper**: Developed goalkeeper actions within a POMDP-based strategy framework using Q-learning in C#, controlled the interception arm, and co-designed the ball-gripping mechanism with ToF possession detection.
- **FIPATECH – French Robotics Cup** Brest
ROS 2/C++ Developer – 17th/89 in the Senior category, 2026 2025–2026
 - **Strategy**: Developed the robot’s real-time planner: online Orienteering Problem solver under time, capacity, and opponent constraints using bounded tree search and Pareto pruning; replanning using observed Nav2 traversal costs.
 - **Radar odometry**: Designed wheel-free 2D odometry using 4 NU60 Doppler radars aimed at the ceiling (DVL-inspired): vx/vy estimation by SVD, RANSAC + IRLS/Huber outlier rejection, bias calibration, and ROS 2 integration at 20 Hz.
- **Challenges**: Aqua.Bot 2024 – ROS 2/Gazebo/Nav2 USV, control and potential-field obstacle avoidance (4th/20); SwarmZ 2024 – drone swarms (4th/16); Dassault UAV Challenge 2023 – maritime rescue *tilt-rotor* VTOL (5th/20).

SELECTED ROBOTICS PROJECTS

- **Robotic fish**: Complete mechatronic design of an autonomous robotic fish: CAD, ESP32/FreeRTOS, caudal propulsion, obstacle avoidance, and UART communication.
- **BlueROV**: Autonomous underwater target tracking using ROS 2 visual servoing (COSMER).

SKILLS

- **Programming languages**: Python, C/C++, C#, TypeScript, Rust, Java, SQL, HTML/CSS
- **Robotics & simulation**: ROS/ROS 2, Nav2, Gazebo, MATLAB/Simulink, OpenCV, PyTorch
- **Tools & CAD**: Linux, Docker, Git, SolidWorks, Inventor, Catia V5, KiCad
- **Languages**: French, English C1 (TOEIC 975/990), German C1 (Abitur 1.4), Chinese A2, Japanese A1

AWARDS AND LEADERSHIP

- **CROUS Short Story Competition**: 2025 winner, 2nd prize in 2024, and jury member in 2026
- **President – Student Association** (2022–2023): Led an association of 200+ members, organized events and managed budgets.